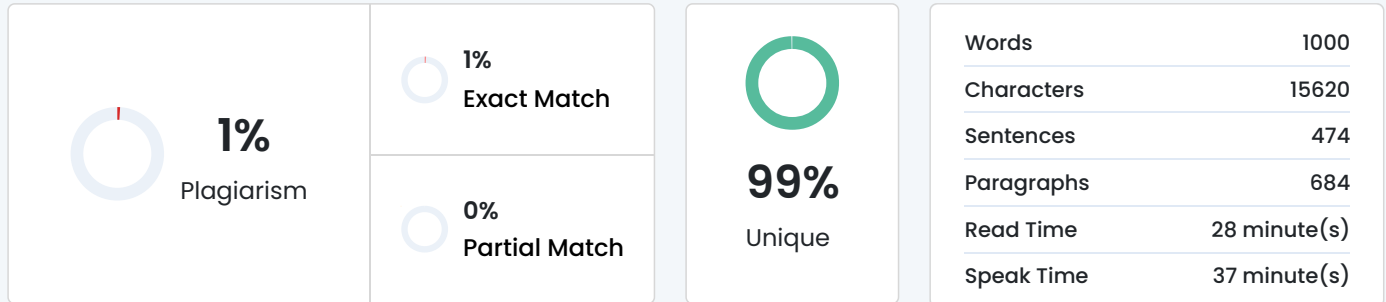


Plagiarism Scan Report



Content Checked For Plagiarism

SAVEETHA UNIVERSITY

Department of Computer Science And Engineering

COMMON COURSE ASSIGNMENT REPORT

Smart Hospital Patient Appointment , Pharmacy Inventory and Alert Notification

System using Java

Course Code & Name: CSA0911 - Programming in Java

Submitted by Team :

1. Ajay S (192424078) - Module 1 Lead
2. Dhilipan T (192421290) - Module 2 Lead
3. Sreevardhan M (192472405) - Module 3 Lead
4. Bhavesh M (192572024) - Module 4 Lead
5. Siddi Rishwanth O (192472148) - Module 5 Lead

Faculty: Dr. Thalpathi Rajasekaran R

A. Assignment Information

Department Computer Science and Engineering

Programme Artificial Intelligence & Data Science

Course Code & Course Name CSA0911 - Programming in Java

Academic Year / Batch 2024-2028

Faculty Name Dr. Thalpathi Rajasekaran

Assignment Title Smart Hospital Patient Appointment , Pharmacy Inventory and Alert Notification

System using Java

Maximum Marks 100

Course Outcome(s) – CO

CO1: OOP principles (classes, encapsulation, inheritance, polymorphism). CO2:

Collection Framework (List, ArrayList, Set, Map, HashMap, Hashtable) with

Iterator/ListIterator and generics. CO3: Multithreading (priorities, synchronization, inter-thread communication) and exception handling.

Bloom's Taxonomy Level L2 – Understand; L3 – Apply; L4 – Analyse

SDG Mapping (SDG 1–17) SDG 3 – Good Health and Well-being; SDG 9 – Industry, Innovation and Infrastructure; SDG 10 – Reduced Inequalities

Industry / Societal Relevance

Digitises hospital front-office and pharmacy workflows that are still largely

paper-based in many small and mid-sized clinics, reducing appointment conflicts,

medicine stock-outs and missed follow-ups.

Table of Contents

A. Assignment Information.....	1
Table of Contents.....	1
1. Problem Statement and Problem Formulation.....	1
1.1 Problem Statement.....	1
1.2 Problem Formulation.....	1
2. Objective and Expected Outcomes.....	1
2.1 Objectives.....	1
2.2 Expected Outcomes.....	1
3. Requirements, Constraints and Assumptions.....	1
3.1 Functional Requirements.....	1
3.2 Non-Functional / Technical Constraints.....	1
3.3 Assumptions.....	1
4. Application of Relevant Course Knowledge / Concepts.....	1
4.1 CO1 – OOP: Classes, Encapsulation, Inheritance, Polymorphism.....	1
4.2 CO2 – Collection Framework, Generics, Iterators.....	1
4.3 CO3 – Exception Handling, Multithreading, Synchronization.....	1
5. Design / Proposed Solution / Methodology.....	1
5.1 Architectural Overview.....	1
5.2 Class Design Summary.....	1
5.3 Key Design Decisions and Alternatives Considered.....	1
6. Algorithm / Pseudocode / Flowchart.....	1
6.1 Flowchart – Book Appointment (described).....	1
7. Implementation / Source Code and Environment / Tools Used.....	1
7.1 Development Environment.....	1
7.2 Representative Source Code.....	1
7.2.1 Module 1 – Polymorphic factory (Medicine).....	1
7.2.2 Module 2 – Collections-backed booking (HospitalService).....	1
7.2.3 Module 3 – Exception-safe menu dispatch (ConsoleApp).....	1
7.2.4 Module 4 – Thread-safe stock reduction (Inventory + JDBC).....	1
7.2.5 Module 5 – Menu loop and Applet action dispatch.....	1
8. Test Cases and Expected/Actual Results.....	1
9. Execution Screenshots / Outputs.....	1
10. Results and Validation.....	1
10.1 Functional Coverage.....	1
10.2 Observations.....	1
11. Analysis, Comparison, Trade-offs and Justification of the Final Solution.....	1
11.1 Design Alternatives Considered.....	1
11.2 Trade-offs of the Final Solution.....	1
11.3 Justification.....	1
12. Broader Considerations / SDG Relevance.....	1
12.1 SDG 3 – Good Health and Well-being.....	1

12.2 SDG 9 – Industry, Innovation and Infrastructure.....	1
12.3 SDG 10 – Reduced Inequalities.....	1
12.4 Ethics, Safety and Professional Responsibility.....	1
13. Conclusion, Limitations and Possible Improvements.....	1
13.1 Conclusion.....	1
13.2 Limitations.....	1
13.3 Possible Improvements.....	1
14. Individual Contribution of Group Members.....	1
Member 1 – Core entity classes and inheritance hierarchy.....	1
Member 2 – The central service facade and its collections.....	1
Member 3 – Custom exceptions and the notification class hierarchy.....	1
Member 4 – Concurrency primitives and the database access layer.....	1
Member 5 – Console and Applet front ends, entry point, overall test coordination.....	1
15. References.....	1
16. One-Page Individual Reflection.....	1
16.1 Individual Reflection – Member 1.....	1
Design/Development Decisions.....	1
Challenges Faced.....	1
Learning Gained.....	1
SDG Connection.....	1
Course Outcome Attainment.....	1
16.2 Individual Reflection – Member 2.....	1
Design/Development Decisions.....	1
Challenges Faced.....	1
Learning Gained.....	1
SDG Connection.....	1
Course Outcome Attainment.....	1
16.3 Individual Reflection – Member 3.....	1
Design/Development Decisions.....	1
Challenges Faced.....	1
Learning Gained.....	1
SDG Connection.....	1
Course Outcome Attainment.....	1
16.4 Individual Reflection – Member 4.....	1
Design/Development Decisions.....	1
Challenges Faced.....	1
Learning Gained.....	1
SDG Connection.....	1
Course Outcome Attainment.....	1
16.5 Individual Reflection – Member 5.....	1
Design/Development Decisions.....	1
Challenges Faced.....	1
Learning Gained.....	1
SDG Connection.....	1
Course Outcome Attainment.....	1

1. Problem Statement and Problem Formulation

1.1 Problem Statement

Hospitals and clinics need to manage patient records, doctor details, appointments, and pharmacy stock efficiently. Manual handling can result in duplicate appointments, difficulty in finding records, incorrect stock information, and delayed alerts for appointments or low medicine availability. The Smart Hospital Patient Appointment, Pharmacy Inventory and Alert Notification System is developed as a menu-driven Java application to simplify these activities by providing patient and doctor management, appointment booking and cancellation, medicine inventory management, stock validation, notifications, and basic report generation.

1.2 Problem Formulation

The system focuses on the following core requirements:

- Entities: Patient, Doctor, Appointment, Medicine, Inventory, and Notification.
- Core workflow: Register → Search/View → Book/Cancel Appointment → Manage Medicine Stock → Generate Reports → Send Notifications.
- Validation: Handle invalid IDs, duplicate appointments, invalid inputs, and insufficient medicine stock using exception handling.
- Collections: Manage hospital data using ArrayList, HashMap, HashSet, Hashtable, generics, Iterator, and ListIterator.
- Concurrency: Demonstrate concurrent appointment booking and background notification processing using threads, synchronization, and inter-thread communication.
- Interfaces and persistence: Provide a console-based menu, a simple AWT GUI, file-based data storage, serialization, and SHA-256 hashing.

The system is designed to demonstrate the practical application of Java programming concepts while keeping hospital appointment and pharmacy operations simple, organized, and reliable.

2. Objective and Expected Outcomes

2.1 Objectives

The primary objective of this project is to develop a Smart Hospital Patient Appointment, Pharmacy Inventory and Alert Notification System using Java that can organize essential hospital and pharmacy activities through a simple menu-driven application. The project aims to apply Java programming concepts to a practical healthcare-management scenario while keeping the system easy to operate and maintain.

The major objectives are:

- To provide a structured system for registering, viewing, and searching patient records.
- To maintain doctor information and appointment details in an organized

manner.

-

Matched Source

Similarity 1%

Title: [What is SCED? - SCED Finder | Fast access to school codes](#)

SCED Course Code is a five-digit code that provides a standardized number for the general subject area and specific subject of the course.

https://google.com/goto?url=CAESZQHrOzAVA7a7v2aJeJfiAUbZJuhDMqvBfEl7yta0N6dOsszCiS_OhtxTYSzNWTnK4h1lyzbbwk3Drf0vK8hvDIWb9yQ6QAeBIlwnYcqVzBFbKBfAYD-Is3gLzLsv0UvyfbEzOdvL

Check By:  Dupli Checker